

Forecasting Oil and Gold Prices with Regime and Structural Demand Features

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Abstract—This study investigates one-step-ahead forecasting of WTI crude oil and gold prices over 2018–2025, with out-of-sample evaluation from 2022 to mid-2025. Linear Regression, XGBoost, and Random Forest are compared using feature sets that include cross-market signals, regime indicators (VIX, GPR, Bitcoin volatility), and global EV adoption as a structural demand proxy. Both log-return and price-level targets are examined. While return-based modeling is commonly used to address non-stationarity, it yields limited practical gains. In contrast, models trained on price levels provide more stable and interpretable performance. XGBoost achieves the lowest errors, with price-level MAPE of 1.82% for oil and 0.74% for gold. EV adoption does not improve short-horizon prediction. Errors increase under higher market stress, and residual analysis indicates that remaining volatility dynamics are better captured by GARCH filtering.

Index Terms—commodity price forecasting, XGBoost, Random Forest, electric vehicles, oil, gold

I. INTRODUCTION

Crude oil and gold are two of the most economically significant commodities, playing distinct roles in global markets. Crude oil is a key driver of industrial activity and energy production, while gold serves as a safe-haven asset and hedge against market uncertainty. Accurate forecasting of their prices is therefore essential for investors, policymakers, and risk managers operating in volatile environments.

Traditional econometric approaches such as ARIMA and linear regression have been widely used due to their interpretability. However, commodity prices exhibit nonlinear and complex dynamics driven by macroeconomic and financial factors, as well as cross-market interactions [1]. These dynamics are further characterized by non-stationarity and volatility effects commonly observed in financial time series, including market sentiment indicators such as VIX and GPR.

Tree-based ensemble methods like XGBoost and Random Forest are well-suited to capturing nonlinear relationships and

complex feature interactions in financial data [2]. However, their effectiveness is highly dependent on the statistical properties of the series being modeled. Financial prices tend to drift over time and shift their distributional properties between sample periods, which makes it hard for any model trained on historical data to generalize cleanly to the future. Forecasting raw price levels directly tends to suffer from this problem. Expressing the outcome as a log-return instead produces a series that is much closer to stationary, which keeps the training and testing environments comparable and makes one-step-ahead prediction more reliable.

Beyond financial variables, the global transition toward electric vehicles represents a structural force reshaping long-run oil demand, as expanding EV adoption displaces a growing share of transportation fuel consumption [3]. However, existing machine learning studies largely treat EV adoption as an outcome rather than a predictor, focusing on modeling its diffusion rather than its potential role in commodity price formation [4]. At the same time, oil price forecasting models predominantly rely on financial and macroeconomic indicators, with limited attention given to slow-moving structural demand variables [5].

This gap raises an important question: whether a structural demand signal, such as EV adoption, is incorporated into short-horizon price dynamics or instead operates only at longer time scales. To address this, the present study incorporates global EV sales as a structural covariate and explicitly evaluates its incremental contribution within a high-frequency forecasting framework. In doing so, the analysis also provides insight into the alignment between feature time scale and prediction horizon in machine learning-based commodity forecasting.

This paper makes three main contributions to the commodity forecasting literature. First, it provides an empirical evaluation

of global EV adoption as a structural demand covariate and shows that, despite its economic relevance, it does not yield meaningful predictive gains at short horizons, highlighting the importance of aligning feature time scales with forecasting objectives. Second, it examines the role of target representation by comparing return-based and price-based modeling approaches, demonstrating that models trained directly on price levels provide more stable and interpretable predictive performance in this setting. Third, it introduces regime-based variables derived from VIX, GPR, and realized volatility to assess model performance under varying market conditions, showing that forecast errors increase systematically under elevated market stress and offering a conditional perspective on forecasting risk.

II. RELATED WORKS

A. Machine Learning for Commodity Price Forecasting

The application of machine learning to commodity price forecasting has grown substantially over the past decade. Prior studies show that tree-based ensemble models provide strong predictive performance in financial time series. In particular, XGBoost has been shown to outperform traditional regression models in crude oil price prediction, highlighting the importance of modeling nonlinear relationships in oil price dynamics [1] [2].

In the context of gold, XGBoost combined with SHAP interaction values improves both predictive accuracy and the interpretability of macroeconomic drivers [6]. Comparative studies further indicate that, despite the flexibility of deep learning architectures, tree-based ensemble methods remain highly competitive for WTI price forecasting [7]. More recent work also demonstrates that combining models in a meta-learning framework can further enhance performance, with volatility indices such as VIX and OVX and lagged oil prices identified as key predictors of short-term price movements [2].

B. Return-Based Modeling and Distribution Shift

A recurring challenge in machine learning-based commodity forecasting is the non-stationarity of raw price levels, which can lead to distribution shifts between training and testing periods. To address this issue, many studies adopt return-based representations, as expressing the dependent variable in returns helps capture short-term dynamics and volatility in oil price movements [2]. Under this framework, models are trained to predict one-step-ahead returns, which can subsequently be transformed back into price levels. This approach improves model robustness by aligning the training and testing distributions and reducing sensitivity to long-term price trends.

C. Role of Volatility Indices and Geopolitical Risk

The role of market-wide volatility in shaping oil price dynamics has been widely documented. Recent studies examine the role of volatility-related variables, including the VIX and realized volatility, in machine learning-based forecasting frameworks, showing that these indicators are consistently identified as important drivers of volatility dynamics [8]. In

parallel, Caldara and Iacoviello construct the widely used Geopolitical Risk (GPR) index, which captures geopolitical tensions and is associated with adverse economic conditions and heightened financial market volatility [9].

Recent machine learning studies show that geopolitical risk components contribute incremental predictive information for crude oil futures volatility, with their impact exhibiting time-varying and asymmetric patterns across economic conditions [10]. However, existing evidence suggests that while volatility indices and risk measures provide valuable information, their predictive contributions are often moderate and context-dependent.

D. Electric Vehicles and Structural Demand Shifts

The global uptake of electric vehicles has become increasingly difficult to ignore as a long-term force bearing on oil consumption. According to industry data, annual EV sales surpassed ten million units in 2022 and have continued on an upward trajectory, with the cumulative reduction in transport fuel demand projected to account for a meaningful share of daily global oil consumption before the end of the decade [3]. Because this transition unfolds over years rather than days, it behaves very differently from the kind of shock that moves oil prices overnight — which is exactly why it might be better thought of as a background structural variable rather than a near-term price driver. Machine learning studies in this space typically frame adoption itself as the variable to be predicted, drawing on inputs such as macroeconomic conditions, vehicle attributes, and consumer behavior data to model how penetration rates evolve across time and geography [4].

III. METHODOLOGY

A. Data Sources

This study employs a multi-source financial and risk dataset covering the period from January 1, 2018, to June 30, 2025. Daily market data for Bitcoin (BTC-USD), the S&P 500 Index (*GSPC), crude oil futures (CL=F), gold futures (GC=F), and the CBOE Volatility Index (*VIX) are obtained from Yahoo Finance [11]. In parallel, the Geopolitical Risk (GPR) index is sourced from the database developed by Matteo Iacoviello [12]. While electric vehicle (EV) sales data are drawn from the *Global EV Outlook 2025* published by the International Energy Agency [3]. EV data are originally available at an annual frequency. The sample is split chronologically into two periods. Training data run from mid-February 2018 through the end of 2021, a period largely free from the major geopolitical disruptions that followed. The test set picks up from January 2022 through late June 2025, covering the period after the outbreak of the Russia-Ukraine war, when commodity markets were operating under substantially elevated uncertainty.

B. Data Preprocessing

Before model estimation, each dataset undergoes a cleaning procedure to ensure consistency across sources. Variable names are standardized, the date column is identified and

converted into a common format, and all non-date variables are cast into numeric form. The observations are restricted to the sample period between January 1, 2018, and June 30, 2025. All variables are then aligned by date and merged into a unified dataset. To maintain data integrity, only observations with complete values across the main variables are retained.

C. Feature Engineering

To enhance the predictive capability of the models, several features are constructed from the preprocessed data. Logarithmic returns for all price series are computed to capture short-term market dynamics. The return is defined as

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

where P_t denotes the asset price at time t . Based on these returns, realized volatility measures are constructed at one-day, seven-day, fifteen-day, and thirty-day horizons using rolling sums of squared returns. Among these, the thirty-day realized volatility (RV30) is selected as the primary volatility feature, as it exhibits the most persistent autocorrelation compared to shorter windows, indicating a more stable representation of volatility dynamics.

Alongside market-based variables, a structural demand proxy is introduced through global EV adoption. Because EV sales are reported at an annual frequency, they are converted into a daily series using linear interpolation between yearly observations, followed by forward filling to maintain continuity. The transformed series is then expressed as $\log(\text{EV} + 1)$ to reduce skewness and reflect the gradual and cumulative nature of the underlying demand shift.

It is important to note that this variable evolves at a much slower temporal scale than financial market variables. Its inclusion is therefore intended not only to test its predictive contribution but also to assess whether slow-moving structural information can be meaningfully incorporated into short-horizon forecasting models.

D. Regime-Based Feature Construction

Three independent regime dimensions are constructed using quantile-based cutoffs estimated exclusively on the training data to avoid look-ahead bias. Bitcoin RV30 is used to define three volatility regimes based on quantile thresholds: low (below the 50th percentile), medium (50th–90th percentile), and high (above the 90th percentile), denoted as S1, S2, and S3, respectively. Analogous regimes are constructed for the GPR index (G1/G2/G3) and the VIX level (V1/V2/V3) using the same quantile-based partitioning.

These regime indicators are encoded as dummy variables and incorporated into regime-augmented model specifications. In addition, they are used to partition the test set for condition-specific evaluation of model accuracy. Alternative threshold configurations were also evaluated, yielding no material differences in performance; therefore, the representative quantile specification described above is adopted throughout.

E. Model Specification

To evaluate predictive performance under different modeling assumptions, a range of statistical and machine learning models is employed, including linear regression (LR) as a benchmark and tree-based ensemble methods such as Random Forest (RF) and Extreme Gradient Boosting (XGBoost) to capture nonlinear interactions within financial time series data. For XGBoost, model performance is optimized through hyperparameter tuning using grid search with cross-validation [6]. The final configuration employs shallow trees (maximum depth = 3), a learning rate of 0.05, 500 boosting rounds, and L2 regularization $\lambda = 1$ to control model complexity and prevent overfitting. Early stopping is applied based on validation performance to avoid excessive training iterations. The Random Forest model is implemented with a fixed number of trees and a controlled random seed to ensure reproducibility [2].

Cross-market variables, particularly S&P 500 returns, are included as predictors to capture potential information spillovers between equity and commodity markets. Five model variants are constructed, each progressively incorporating additional information. The baseline specification includes own-asset returns, volatility, and broad market indicators. This is followed by the inclusion of cross-asset returns and volatilities to account for inter-market linkages, then regime indicators, and finally the EV variable. This sequential design enables a systematic assessment of the incremental contribution of each feature group.

The prediction target is defined as the one-step-ahead log-return y_{t+1} , as introduced in (1). Given the predicted return $\hat{y}_{t+1}$, the corresponding price forecast is obtained as

$$\hat{P}_{t+1} = P_t \cdot \exp(\hat{y}_{t+1}). \quad (2)$$

This formulation enables evaluation at both the return level and the price level using metrics such as RMSE, MAE, and MAPE.

F. Evaluation Metrics

Model performance is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), which are widely used metrics in model evaluation [13]. MAE measures the average magnitude of prediction errors, while RMSE penalizes larger errors more heavily due to the squaring of residuals.

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (4)$$

The Mean Absolute Percentage Error (MAPE) is not suitable for return predictions, as log-returns often take values close to zero. This causes percentage-based errors to become unstable or undefined. In contrast, MAPE is appropriate for price predictions since price levels are strictly positive and avoid this issue.

IV. RESULTS AND DISCUSSION

A. Cross-Market Relationships and Information Flow

Pairwise correlations indicate moderate positive comovement between S&P 500 and oil returns, consistent with global risk-on/risk-off dynamics. In contrast, oil and gold exhibit weaker correlation, reflecting gold's partial independence from cyclical economic activity. Bitcoin shows the lowest correlation with both assets, suggesting that it carries distinct information relative to traditional financial markets.

Granger causality tests, reported in Table I, reveal a clear directional relationship. S&P 500 returns significantly Granger-cause oil returns, confirming that equity market information flows into commodity pricing with a measurable lead. In contrast, reverse causality is not observed, indicating a unidirectional information channel. The relationship between oil and gold is weaker and only marginally significant, while gold does not Granger-cause oil. These findings provide empirical support for the inclusion of cross-market variables, particularly S&P 500 returns, in the predictive modeling framework.

TABLE I
GRANGER CAUSALITY TEST RESULTS (LAG ORDER = 2)

Source	Target	F-Stat	p-value	Result
S&P 500	Oil	12.858	2.86×10^{-6}	Significant
Oil	S&P 500	0.0725	0.9301	Not Significant
Oil	Gold	2.5297	0.0800	Marginally Significant
Gold	Oil	0.7457	0.4746	Not Significant

B. Return Prediction Performance

Table II reports the out-of-sample return prediction performance across all model specifications. Overall, XGBoost consistently achieves the lowest RMSE and MAE for both oil and gold, indicating that nonlinear modeling is more effective than linear regression and Random Forest in capturing return dynamics. This pattern aligns with prior findings that tree-based methods outperform traditional models in oil price prediction, reflecting their ability to capture the nonlinear and complex dynamics inherent in commodity markets [1]. The mean return is also reported to provide a scale reference for interpreting the magnitude of prediction errors. Notably, oil returns are centered close to zero, which helps explain the relatively large error values in absolute terms.

For oil, the performance gap between XGBoost and the other models is more pronounced, suggesting stronger nonlinearities in oil return behavior. The inclusion of EV-related features does not lead to a consistent improvement in predictive performance across models. For oil, the impact is negligible for XGBoost and modestly adverse for Random Forest, while for gold, the effect is statistically indistinguishable from zero. These results suggest that, at a daily forecasting horizon, structural demand signals captured by EV adoption do not provide economically meaningful incremental information beyond that already contained in market-based variables.

This finding is consistent with the interpretation that EV adoption operates at a lower-frequency structural level and that

its effects on oil demand are incorporated into prices gradually rather than through short-term return fluctuations.

TABLE II
OUT-OF-SAMPLE RETURN PREDICTION PERFORMANCE

Asset	Model	RMSE	MAE	Mean Return
Oil	LR	0.024580	0.018832	-0.000164
Oil	LR + EV	0.024421	0.018578	-0.000164
Oil	LR + EV + Regime	0.024546	0.018579	-0.000164
Oil	XGB	0.023668	0.018180	-0.000164
Oil	XGB + EV	0.023693	0.018194	-0.000164
Oil	XGB + EV + Regime	0.023638	0.018148	-0.000164
Oil	RF	0.026575	0.020228	-0.000164
Oil	RF + EV	0.027110	0.021225	-0.000164
Oil	RF + EV + Regime	0.027635	0.021527	-0.000164
Gold	LR	0.009922	0.007450	0.000733
Gold	LR + EV	0.009921	0.007450	0.000733
Gold	LR + EV + Regime	0.010073	0.007531	0.000733
Gold	XGB	0.009814	0.007376	0.000733
Gold	XGB + EV	0.009814	0.007378	0.000733
Gold	XGB + EV + Regime	0.009833	0.007381	0.000733
Gold	RF	0.010004	0.007497	0.000733
Gold	RF + EV	0.009938	0.007432	0.000733
Gold	RF + EV + Regime	0.009946	0.007452	0.000733

C. Price Reconstruction Performance

Table III reports the out-of-sample performance evaluated at the price level. Compared to return-based evaluation, differences across models become more pronounced, indicating that price-based metrics provide a more discriminative assessment of practical forecasting performance.

XGBoost again achieves the best overall performance, particularly for oil. At the price level, the inclusion of EV variables yields marginal and non-uniform changes in predictive accuracy across model classes. While slight improvements are observed in certain linear specifications, tree-based models show either negligible differences or mild deterioration. Taken together, these results indicate that EV-related information does not systematically enhance short-horizon price prediction.

Rather than reflecting a lack of economic relevance, this pattern highlights a scale mismatch between the slow evolution of structural demand factors and the high-frequency dynamics of commodity price movements [5]. As such, EV adoption appears to function as a background structural variable whose influence is more likely to manifest over longer forecasting horizons.

Adding regime dummies does not produce any consistent improvement in price-level accuracy either. Once continuous volatility indicators like VIX are already in the model, the categorical regime labels seem to carry little additional information — the model has already absorbed what the regimes are trying to represent. For gold, the story across all models is straightforward, MAPE hovers between 0.74% and 0.76% regardless of specification, reflecting the fact that gold price movements are relatively smooth and do not demand much nonlinear modeling power to track reasonably well.

D. Feature Importance Analysis

Figure 1 shows the top predictors ranked by XGBoost information gain for the oil and gold models. For oil, S&P 500

TABLE III
OUT-OF-SAMPLE PRICE PREDICTION PERFORMANCE

Asset	Model	RMSE	MAE	MAPE (%)
Oil	LR	2.1654	1.5731	1.8864
Oil	LR + EV	2.1507	1.5531	1.8689
Oil	LR + EV + Regime	2.1639	1.5531	1.8683
Oil	XGB	2.0864	1.5152	1.8246
Oil	XGB + EV	2.0890	1.5172	1.8272
Oil	XGB + EV + Regime	2.0883	1.5164	1.8258
Oil	RF	2.2984	1.6733	2.0116
Oil	RF + EV	2.3484	1.7606	2.1168
Oil	RF + EV + Regime	2.3991	1.7864	2.1449
Gold	LR	23.7430	16.7310	0.7478
Gold	LR + EV	23.7409	16.7290	0.7478
Gold	LR + EV + Regime	24.1477	16.9160	0.7560
Gold	XGB	23.5262	16.5719	0.7413
Gold	XGB + EV	23.5014	16.5510	0.7409
Gold	XGB + EV + Regime	23.4817	16.5499	0.7405
Gold	RF	23.7424	16.7684	0.7531
Gold	RF + EV	23.6768	16.6263	0.7454
Gold	RF + EV + Regime	23.6544	16.6617	0.7475

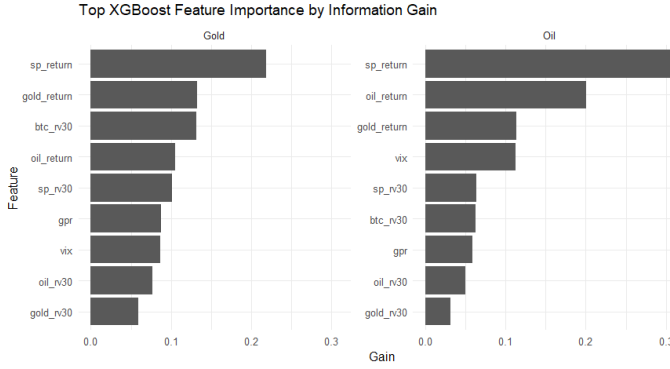


Fig. 1. Top predictors in the XGBoost models ranked by information gain

return is the most important feature, followed by own-asset return and gold return, highlighting the role of both momentum and cross-market linkages. VIX and realized volatility measures also contribute meaningfully, indicating the importance of uncertainty in oil price dynamics.

For gold, the S&P 500 return again ranks first, followed by the gold return and Bitcoin volatility. This suggests that gold is influenced more by broad financial sentiment and cross-market risk conditions than by its own momentum alone. Overall, the rankings indicate that oil and gold respond to different underlying drivers.

E. Regime-Based Analysis

The regime-based analysis is conducted across three dimensions: realized volatility (RV30), geopolitical risk (GPR), and market fear (VIX). Among these, the VIX-based regime provides the clearest separation in predictive performance. As shown in Fig. 2, prediction errors increase monotonically from low to high stress levels for both oil and gold under the VIX regime. This effect is particularly pronounced for oil, where RMSE rises sharply under elevated market stress, indicating a substantial deterioration in model accuracy [10].

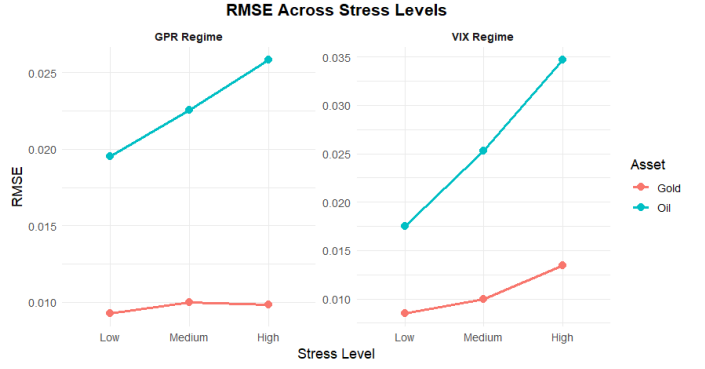


Fig. 2. RMSE across stress levels for VIX and GPR regimes

A similar but less pronounced pattern is observed under the GPR regime, where prediction errors also increase with higher risk levels; although the magnitude of change is smaller compared to VIX. For practitioners, this has a direct operational implication. Rather than relying on regime indicators to sharpen point forecasts, the more appropriate use of these variables is as a real-time signal for widening uncertainty bounds and reducing exposure during periods when market stress metrics climb above historical norms. In contrast, the volatility-based regime (RV30) does not provide a complete comparison, as no observations fall into the high-volatility category in the test sample. Consequently, it is excluded from the low–high comparative analysis.

F. Residual Analysis and Model Behavior

Residual diagnostics provide further insight into model adequacy. Shapiro–Wilk tests indicate that residuals are non-normal across all models ($p < 0.01$), which is consistent with the heavy-tailed nature of financial return distributions. Ljung–Box tests reveal a clear difference between assets. For oil, several model specifications exhibit significant residual autocorrelation, indicating that temporal dependencies remain unmodeled. In contrast, no significant autocorrelation is detected for gold, suggesting that the models adequately capture its return dynamics.

To address the remaining dependence in oil residuals, ARIMA-based residual modeling is first applied. However, this approach fails to eliminate autocorrelation and yields negligible improvements in RMSE and MAE, indicating that the issue does not originate from linear mean dynamics. Subsequently, a GARCH(1,1) model is fitted to the residuals. This significantly reduces autocorrelation for the XGBoost model, confirming the presence of volatility clustering. For the Random Forest model, autocorrelation is only partially reduced, with significant dependence remaining in some lags, indicating limited improvement. These findings suggest that the remaining structure in the residuals is primarily driven by time-varying variance rather than the conditional mean. It should be noted that the GARCH(1,1) correction here serves as a diagnostic exercise applied to model residuals after forecasting is com-

plete, rather than as a component of an integrated prediction pipeline.

TABLE IV
RESIDUAL DIAGNOSTIC SUMMARY FOR OIL MODELS

Model	Initial Ljung-Box	After ARIMA	After GARCH
LR	Detected	Detected	Detected
XGB	Detected	Detected	Not detected
RF	Detected	Detected	Limited improvement

V. CONCLUSION

This study evaluated a machine learning framework for one-step-ahead forecasting of WTI crude oil and gold futures returns and prices from 2022 through mid-2025, using training data from 2018–2021. Three model families, Linear Regression, XGBoost, and Random Forest, were tested across five feature configurations that progressively introduced cross-market signals, regime indicators derived from VIX, GPR, and Bitcoin realized volatility, and global EV adoption as a structural covariate. Across all specifications and both assets, XGBoost consistently delivered the lowest out-of-sample errors, achieving an RMSE of 0.02367 and MAE of 0.01815 for oil returns, and 0.00981 and 0.00738 for gold, confirming the advantage of nonlinear ensemble methods in capturing complex commodity dynamics. At the price level, XGBoost reached a MAPE of 1.82% for oil and 0.74% for gold.

The central question of whether global EV adoption contributes to short-horizon commodity price prediction is answered in the negative. Incorporating EV data produces only marginal and inconsistent changes in predictive performance across all model classes. This null result is nevertheless informative. It highlights a fundamental temporal misalignment: while EV adoption is economically relevant as a driver of long-run oil demand, it does not translate into a usable predictive signal in daily return models.

This finding suggests that structural energy transition variables are incorporated into commodity prices over longer horizons and are not directly exploitable for high-frequency forecasting. More broadly, it underscores the importance of aligning feature time scales with prediction horizons when designing machine learning models for financial applications.

Regime-based analysis shows that forecast errors rise monotonically with market stress under both VIX and GPR classifications, with the sharpest deterioration observed for oil. While regime dummies do not materially improve aggregate accuracy metrics, they provide a meaningful conditional risk lens — particularly useful for widening uncertainty bounds during elevated-stress periods rather than sharpening point forecasts. Residual diagnostics further reveal that oil model residuals carry significant autocorrelation that ARIMA correction fails to resolve, but a GARCH(1,1) filter applied to XGBoost residuals successfully eliminates it, confirming that the remaining structure is driven by volatility clustering rather than linear mean dependence.

Future work should consider integrating machine learning methods with volatility models in a unified pipeline rather

than as a post-hoc diagnostic to operationalize the gains suggested by the GARCH results. Higher-frequency intraday data and sequence-based architectures may better capture long-run temporal dependencies, while replacing historical EV sales with forward-looking penetration projections could produce a demand covariate that more directly reflects market expectations regarding the energy transition.

REFERENCES

- [1] H. Guliyev and E. Mustafayev, "Predicting the changes in the WTI crude oil price dynamics using machine learning models," *Resources Policy*, vol. 77, p. 102664, 2022, doi: 10.1016/j.resourpol.2022.102664.
- [2] G. Cohen, "A Comprehensive Study on Short-Term Oil Price Forecasting Using Econometric and Machine Learning Techniques," *Machine Learning and Knowledge Extraction*, vol. 7, p. 127, 2025, doi: 10.3390/make7040127.
- [3] International Energy Agency, "Global EV Outlook 2025," [Online]. Available: <https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2025>
- [4] S. Afandizadeh, D. Sharifi, N. Kalantari, and H. Mirzahosseini, "Using machine learning methods to predict electric vehicles penetration in the automotive market," *Scientific Reports*, vol. 13, art. no. 8345, 2023, doi: 10.1038/s41598-023-35366-3.
- [5] J. Luo, T. Klein, T. Walther, and Q. Ji, "Forecasting realized volatility of crude oil futures prices based on machine learning," *Journal of Forecasting*, vol. 43, no. 5, pp. 1422–1446, 2024, doi: 10.1002/for.3077.
- [6] S. B. Jabeur, S. Mefteh-Wali, and J.-L. Viviani, "Forecasting gold price with the XGBoost algorithm and SHAP interaction values," *Annals of Operations Research*, vol. 334, pp. 679–699, 2024, doi: 10.1007/s10479-021-04187-w.
- [7] P. Foroutan and S. Lahmiri, "Deep learning systems for forecasting the prices of crude oil and precious metals," *Financial Innovation*, vol. 10, art. no. 111, 2024, doi: 10.1186/s40854-024-00637-z.
- [8] E. S. Gunnarsson, H. R. Isern, A. Kaloudis, M. Risstad, B. Vigdel, and S. Westgaard, "Prediction of realized volatility and implied volatility indices using AI and machine learning: A review," *International Review of Financial Analysis*, vol. 93, art. no. 103221, 2024, doi: 10.1016/j.irfa.2024.103221.
- [9] D. Caldara and M. Iacoviello, "Measuring Geopolitical Risk," *International Finance Discussion Papers*, no. 1222r1, Board of Governors of the Federal Reserve System, Washington, 2022, doi: 10.17016/IFDP.2022.1222r1.
- [10] H. Zhang, W. Wang, and Z. Niu, "Geopolitical risks and crude oil futures volatility: Evidence from machine learning," *Resources Policy*, vol. 98, art. no. 105374, 2024, doi: 10.1016/j.resourpol.2024.105374.
- [11] Yahoo Finance, "Historical market data," [Online]. Available: <https://finance.yahoo.com/>. [Accessed: Apr. 18, 2026].
- [12] M. Iacoviello, "Geopolitical Risk (GPR) Index," [Online]. Available: <https://www.matteoiacoviello.com/gpr.htm>. [Accessed: Apr. 18, 2026].
- [13] T. O. Hodson, "Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not," *Geoscientific Model Development Discussions*, vol. 15, pp. 5481–5487, 2022, doi: 10.5194/gmd-15-5481-2022.